

Note:

- During the attendance check a sticker containing a unique code will be put on this exam.
- This code contains a unique number that associates this exam with your registration number.
- This number is printed both next to the code and to the signature field in the attendance check list.

Operations Research

Exam: IN0024 / Midterm

Date: Tuesday 12th May, 2026

Examiner: Prof. Dr. Martin Bichler

Time: 08:00 – 09:00

Working instructions

- This exam consists of **12 pages** with a total of **4 problems**.
Please make sure now that you received a complete copy of the exam.
- The total amount of achievable credits in this exam is 60 credits.
- Detaching pages from the exam is prohibited.
- Allowed resources:
 - one **non-programmable pocket calculator**
 - one **analog dictionary** English ↔ native language
 - one **ruler/geo triangle**
- **Answers are only accepted if the solution approach is documented.** Give a reason for each answer unless explicitly stated otherwise in the respective subproblem.
- Do not write with red or green colors nor use pencils.
- Answer the exam questions in English or German.
- Physically turn off all electronic devices, put them into your bag and close the bag.

Left room from _____ to _____ / Early submission at _____

Problem 1 LP Modelling (14 credits)

A small factory produces two products: A and B. Let $x_A \geq 0$ and $x_B \geq 0$ denote the number of units produced for product A and B, respectively.

Each unit of product A requires 2 hours of machine time and 1 unit of raw material. Each unit of product B requires 1 hour of machine time and 2 units of raw material. The factory has at most 100 hours of machine time and 80 units of raw material available per week.

Formulate the following constraints using only linear (in) equalities.



a) The profit per unit is 30 EUR for product A and 40 EUR for product B. Formulate a suitable objective.

$$\max 30x_A + 40x_B$$



b) Formulate the constraint on machine time.

$$2x_A + x_B \leq 100$$



c) Formulate the constraint on raw material.

$$x_A + 2x_B \leq 80$$



d) Due to contractual obligations, at least 10 units of product A must be produced.

$$x_A \geq 10$$



e) At least 60% of all produced units must be product A. Formulate this constraint linearly with integer coefficients in the form $ax_A + bx_B \leq 0$ with $a, b \in \mathbb{Z}$.

$$x_A \geq 0.6(x_A + x_B) \Leftrightarrow 0.4x_A - 0.6x_B \geq 0 \Leftrightarrow -2x_A + 3x_B \leq 0$$

f) $\max\{|17x_A + 5x_B|, |3x_A + 37x_B|\} \leq 42$.

	0
	1
	2
	3

For the max constraints we first note that as x_A and x_B are non-negative and all involved constants are non-negative as well. It follows that the absolute values can be dropped. The constraint $\max_i\{f(x_i) \leq c\}$ can be replaced by the individual constraints $f(x_i) \leq c, \forall i$

g) We now consider an extension to the previous model:
If more than 30 units of A are produced, each additional unit requires 3 hours of machine time instead of 2. Introduce suitable new variables which model this constraint linearly. State the new formulation of any of the previous constraints that is affected by this extension.

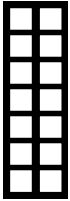
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We introduce $x_{A,L} \geq 0$ and $x_{A,H} \geq 0$ and require $x_A = x_{A,L} + x_{A,H}$, as well as $x_{A,L} \leq 30$. Only the constraint for machine time needs to be updated:

$$2x_{A,L} + 3x_{A,H} + x_B \leq 100.$$

Problem 2 Graphical Solution (17 credits)

0
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6

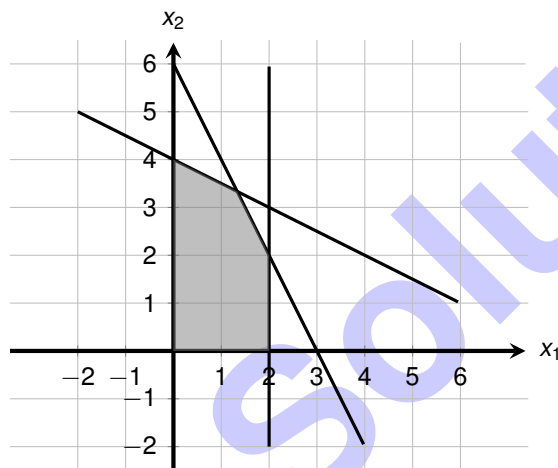


a) Consider the linear program

$$\begin{aligned} \max \quad & 5x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \leq 8 \\ & 2x_1 + x_2 \leq 6 \\ & x_1 \leq 2 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Draw the feasible region and find the optimal solution and optimal value.

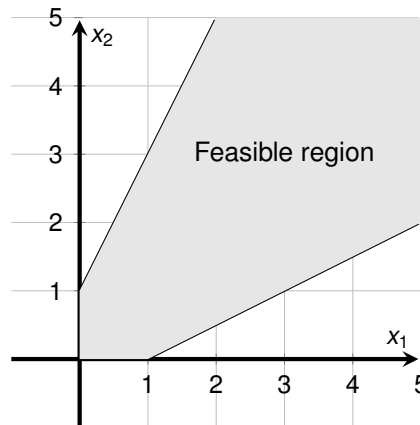
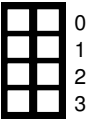
(i) Feasible Region



(ii) Optimal Solution Optimal solution:

$$x^* = (2, 2), \quad z^* = 14.$$

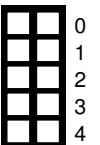
b) Determine a system of linear inequalities describing the unbounded feasible region given by the following shaded area:



From the plot, the region is bounded by:

$$\begin{aligned}x_2 &\leq 1 + 2x_1 \\x_1 &\leq 1 + 2x_2 \\x_1, x_2 &\geq 0.\end{aligned}$$

c) State **all** objective vectors $c = (c_1, c_2) \in \mathbb{R}^2$ such that there are multiple optimal solutions in the feasible region from Part (b). State the corresponding objective values.



The objective vectors are

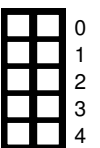
$$(-1, 0), (0, -1), (1, -2), (-2, 1).$$

Their associated objective values are

$$0, 0, 1, 1.$$

d) Characterize **all** objective vectors $c = (c_1, c_2) \in \mathbb{R}^2$ such that the objective $\max c_1 x_1 + c_2 x_2$ is unbounded on the feasible region from Part (b).

Hint: Partial points are awarded if only a subset of vectors is specified that fulfil the condition above.



The normal vectors of the two non-trivial boundary conditions are:

$$n_1 = \begin{pmatrix} 1 \\ -2 \end{pmatrix}, \quad n_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

There are infinitely many optimal solutions for $c = n_i$, all with the same objective value of 1. Thus, the set of objective vectors for which the LP is unbounded is

$$\{\lambda_1 n_1 + \lambda_2 n_2, \lambda_i > 0\}.$$

Problem 3 Simplex Algorithm (16 credits)

0		
1		
2		
3		

a) Transform the following LP into canonical form.

$$\begin{aligned} \max \quad & 5x_1 + 2x_2 \\ \text{s.t.} \quad & -x_1 - 2x_2 \geq -8 \\ & 2x_1 + x_2 \leq 6 \\ & x_1 \leq 2 \\ & x_1, x_2 \geq 0. \end{aligned}$$

Step 1: Convert inequalities to standard form (all $\leq$):

$$-x_1 - 2x_2 \geq -8 \quad \Rightarrow \quad x_1 + 2x_2 \leq 8$$

Step 2: Introduce slack variables $x_3, x_4, x_5 \geq 0$:

The canonical form becomes:

$$\begin{aligned} \max \quad & 5x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 + x_3 = 8, \\ & 2x_1 + x_2 + x_4 = 6, \\ & x_1 + x_5 = 2, \\ & x_1, x_2, x_3, x_4, x_5 \geq 0. \end{aligned}$$

	0
	1
	2
	3
	4
	5
	6

b) Perform **one** iteration of the simplex algorithm. State the incoming and outgoing basic variables clearly.

Step 1: Initial tableau

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	-5	-2	0	0	0	0
x_3	1	2	1	0	0	8
x_4	2	1	0	1	0	6
x_5	1	0	0	0	1	2

Step 2: Choose entering and leaving variables:

Most negative coefficient in z -row: -5 corresponds to $x_1 \rightarrow$ entering variable.

Compute ratios for leaving variable:

$$\frac{b_i}{a_{i,\text{entering}}} = \frac{8}{1} = 8, \quad \frac{6}{2} = 3, \quad \frac{2}{1} = 2$$

Minimum positive ratio: 2 (from x_5) $\Rightarrow x_5$ leaves.

Step 3: Pivot to update tableau (one iteration)

Base	Coefficients					Res.
	x_1	x_2	x_3	x_4	x_5	
z	0	-2	0	0	5	10
x_3	0	2	1	0	-1	6
x_4	0	1	0	1	-2	2
x_1	1	0	0	0	1	2



c) State $x^{(1)} \in \mathbb{R}^2$, the point that the simplex algorithm reaches after one iteration, clearly.

The current basic variables are $x_1 = 2$, $x_3 = 6$, $x_4 = 2$, $x_2 = x_5 = 0$:

$$x^{(1)} = (x_1, x_2) = (2, 0)$$



d) Is the resulting point $x^{(1)}$ optimal? Justify your answer.

No, because the z-row still has a negative coefficient -2 for x_2 . Therefore, the current solution is **not optimal**.



e) Explain degeneracy in the context of the simplex algorithm and state how it can be detected in the simplex tableau.

Definition: Degeneracy occurs when a basic variable takes the value 0 in a basic feasible solution, or alternatively, if multiple redundant constraints intersect at a vertex.

Detection in tableau: If the minimum ratio test yields 0 for leaving variable, the pivot results in no increase in the objective value.

f) Your colleague has computed all vertices of **another** linear program and some of the corresponding associated objective values, specified below.
Determine the optimal solution (assuming a maximization problem) and justify why none of the other vertices, for which the objective value is not specified, is optimal.
Hint: You do not need to compute the missing objective values.

Vertex (x_1, x_2)	$(1, 0)$	$(0, 1)$	$(-1, 0)$	$(0, -1)$	$(\frac{3}{4}, -\frac{3}{4})$	$(\frac{3}{4}, \frac{3}{4})$	$(-\frac{3}{4}, \frac{3}{4})$	$(-\frac{3}{4}, -\frac{3}{4})$
Objective	1	?	-1	1	$\frac{3}{2}$	0	$-\frac{3}{2}$	?

Table 3.1: Table of vertices and their objective values.

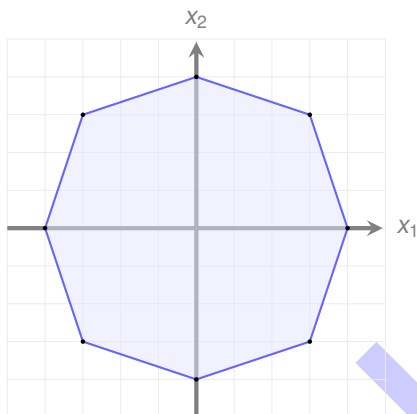


Figure 3.1: Feasible region defined by the given vertices.

The maximum value is $\frac{3}{2}$ at $(\frac{3}{4}, -\frac{3}{4})$.

It suffices to look at the adjacent vertices $(1, 0)$ and $(0, -1)$ with an objective value of $1 < \frac{3}{2}$ to prove optimality.

Problem 4 Sensitivity Analysis (13 credits)

A bakery produces *croissants* (x_1) and *muffins* (x_2) using two limited resources: butter and flour. The following table summarizes the resource consumption and (unknown) profits:

Product	Butter (kg)	Flour (kg)	Profit (EUR)
Croissant (x_1)	2	1	c_1
Muffin (x_2)	1	2	c_2
Capacity	8	10	

The bakery solves the following linear program:

$$\begin{aligned}
 \max \quad & c_1 x_1 + c_2 x_2 \\
 \text{s.t.} \quad & 2x_1 + x_2 \leq 8, \\
 & x_1 + 2x_2 \leq 10, \\
 & x_1, x_2 \geq 0.
 \end{aligned}$$

The optimal simplex tableau is given by:

Base	Coefficients				Res.
	x_1	x_2	s_1	s_2	
Z	0	0	$c_2 - \frac{1}{2}c_1$	$c_1 - \frac{1}{2}c_2$	$4c_1 + 3c_2$
x_1	1	0	$\frac{1}{2}$	$-\frac{1}{2}$	2
x_2	0	1	$-\frac{1}{2}$	$\frac{1}{2}$	4

a) What is the economic interpretation of the slack variables s_1 and s_2 ?

The slack variables represent unused resources:

- s_1 : unused butter (kg)
- s_2 : unused flour (kg)

b) Determine conditions on c_1 and c_2 such that the current basis remains optimal.

For optimality, reduced costs must be non-negative:

$$c_2 - \frac{1}{2}c_1 \geq 0, \quad c_1 - \frac{1}{2}c_2 \geq 0.$$

Equivalently:

$$c_2 \geq \frac{1}{2}c_1, \quad c_1 \geq \frac{1}{2}c_2.$$

For the next tasks, assume $c_1 = 4$ and $c_2 = 3$.

c) What is the shadow price of butter?



Insert $c_1 = 4$, $c_2 = 3$:

$$\text{Shadow price of butter} = c_2 - \frac{1}{2}c_1 = 3 - 2 = 1.$$

d) Should the bakery buy additional butter at a price of 1 EUR/kg? Justify your answer.

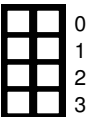


Since butter costs 1 EUR/kg:

$$\text{Net gain} = 1 - 1 = 0.$$

Therefore, the bakery is indifferent (no profit increase).

e) The butter capacity changes from 8 to $8 + \delta$. For which values of δ does the current basis remain feasible?



Let butter capacity change by δ . Use the s_1 column to update RHS:

$$x_1 = 2 + \frac{1}{2}\delta, \quad x_2 = 4 - \frac{1}{2}\delta.$$

Feasibility requires:

$$2 + \frac{1}{2}\delta \geq 0 \Rightarrow \delta \geq -4,$$

$$4 - \frac{1}{2}\delta \geq 0 \Rightarrow \delta \leq 8.$$

Thus:

$$\delta \in [-4, 8].$$

f) Explain why the shadow price is only valid within this range.



Outside this range, one of the basic variables becomes negative, so the current basis is no longer feasible. A new basis must be computed, and thus the shadow price changes.

Additional space for solutions—clearly mark the (sub)problem your answers are related to and strike out invalid solutions.

